

7(a)(i)	$E = V/d$ or $E = F/Q$	C1
	$F = (450 \times 1.60 \times 10^{-19}) / 6.0 \times 10^{-3}$	C1
	$= 1.2 \times 10^{-14} \text{ N}$	A1
	direction: vertically downwards	B1

Question	Answer	Marks
7(a)(ii)	work done = Fs or Fd or EQd	C1
	$= (-)1.2 \times 10^{-14} \times 6.0 \times 10^{-3}$ $= (-)7.2 \times 10^{-17} \text{ J}$	A1
	or	
	work done = VQ	(C1)
	$= (-)450 \times 1.60 \times 10^{-19}$ $= (-)7.2 \times 10^{-17} \text{ J}$	(A1)
7(b)	$E = \frac{1}{2}mv^2$	C1
	$3.4 \times 10^{-16} = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$ $v = 2.7 \times 10^7 \text{ m s}^{-1}$	A1
7(c)(i)	${}^1_1\text{p}$	A1
	${}^0_{-1}\text{e}$	A1
7(c)(ii)	1. hadrons	B1
	2. leptons	B1